PENGUINS: A COMPACT KNOWLEDGE BASE

Habitat and distribution Penguins are flightless seabirds found naturally in the Southern Hemisphere. Their habitats include Antarctica and subantarctic islands, the coasts of South America, southern Africa, Australia, New Zealand, and the Galapagos Islands. Not every penguin lives in an icy climate.

Movement in water Penguin wings evolved into strong, stiff flippers that are suited to swimming rather than flight. In the water, penguins use their flippers like paddles while their feet and tail help with steering. A streamlined body reduces drag, helping them swim quickly and efficiently.

Insulation and feathers Penguins conserve heat with a layer of body fat and densely packed feathers. Air trapped among those feathers adds insulation in cold water. Penguins preen regularly to align their feathers and spread protective oils, which helps maintain a waterproof outer layer.

Camouflage Most penguins have a dark back and a pale belly, a pattern called countershading. From above, the dark back blends with deep water; from below, the pale belly blends with the bright surface. This camouflage can make penguins less visible to predators and prey.

Diet and feeding Penguins mainly eat marine animals such as krill, fish, and squid. Diet differs by species, season, and location: smaller penguins may favor krill and small fish, while larger species can pursue larger fish and squid. Penguins hunt by sight and make repeated dives during feeding trips. Some feed near shore, while others travel many kilometers to productive feeding areas.

Breeding and courtship Many species breed in large colonies where thousands of birds nest close together. Courtship may include calls, postures, and repeated movements that help partners recognize each other. Depending on species and habitat, nests may be built from pebbles or grass, or may simply be shallow scrapes.

Eggs and chicks Emperor penguins do not build a nest for their single egg. The male balances the egg on his feet beneath a warm fold of skin called a brood pouch. In many species, parents share care by alternating between guarding the egg or chick and returning from the sea with food.

Predators At sea, penguins may be hunted by leopard seals, orcas, and sharks. Eggs and chicks can be taken by seabirds such as skuas and giant petrels.

Human pressures Penguins face human-driven threats including overfishing, pollution, habitat disturbance, introduced predators, and accidental capture in fishing gear. Climate-driven

changes in sea ice, ocean temperature, and prey availability can also alter breeding and feeding conditions.

Ocean health Penguins connect land-based breeding colonies with marine food webs. Because their survival and breeding success depend on prey and ocean conditions, changes in penguin populations can serve as indicators of ocean ecosystem health.